

Home Valuation and Two-sided Causal Decisions

Valuation, behavioral response,
and portfolio economics

Valuation $\neq$ purchase offer $\neq$ resale list price

A two-sided decision system built on a valuation distribution:
acquisition offers, seller signals, and upward escalation;
resale listings, buyer signals, and downward markdown;
lifecycle economics, risk, and capital allocation.

Scope: public context, explicit modeling assumptions, and synthetic evidence.

Status: an implemented prototype and proposed extensions, not a description of a company's internal production system.

Snapshot: 2026-09-21.

A concise, standalone synthesis. Maintained data contracts, generated result tables, and source verification limits accompany it in the repository.

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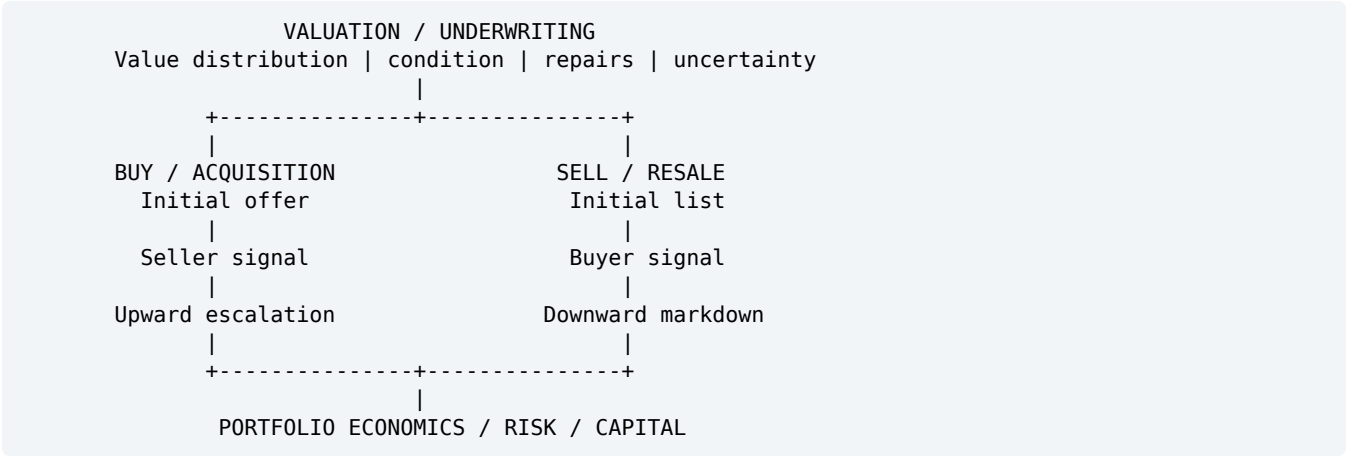
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Technical synthesis

1. Decision architecture

The system is best understood as **two-sided decision-making built on a valuation distribution**, not a single price predictor. Valuation and underwriting describe the asset and its uncertainty. Causal response models describe how feasible actions change homeowner and buyer behavior. The decision layer combines these responses with lifecycle economics, risk, and available capital.

Valuation is not the purchase offer, and neither is the resale list price.



This summary presents the framework, its causal interpretation, and an implemented synthetic prototype. Public information motivates the context; numerical findings are synthetic, not estimates of Opendoor’s actual response curves, internal policy, or operating performance. Maintained field contracts, generated results, and source verification limits accompany this summary in `data_dictionary.md`, `reference_results.md`, `two_sided_results.md`, and `sources.md`.

Quantity	Meaning	Decision implication
Valuation distribution	Possible market/exit values under a stated date, condition, and transaction definition	Foundation for underwriting; not itself a customer quote
Purchase offer	Gross acquisition quote, with seller net proceeds determined by explicit terms	Trades off acquisition probability, selected inventory, cost, and risk
Resale list price	An asking-price action, distinct from eventual transaction proceeds	Trades off buyer response, negotiation, time to close, and retained inventory value

For example, a **stipulated** value distribution centered near \$500,000 can coexist with a \$450,000 acquisition offer and a \$510,000 initial list price. These are not inconsistent valuations: the quotes serve different sides of the business and incorporate costs, risk, competition, and response. They are illustrative numbers, not recommended spreads.

2. Valuation and underwriting as a distribution

Define the target before selecting an algorithm:

$$F_V(v \mid X, t, s) = P(V_{\text{exit}} \leq v \mid X, t, s).$$

X contains point-in-time property and market information, t is the valuation date, and s specifies condition, sale horizon, and market scenario. As-is market value, renovated exit value, and the price from a forced disposition are different targets. A repair cost is not automatically an equal increase in market value.

The model should supply a central estimate and useful uncertainty information, such as calibrated quantiles or scenario distributions. Underwriting also requires forecasts of repairs, holding time, selling expense, and financing. Their errors can be correlated: a market decline can simultaneously reduce exit value, slow completion, and increase capital usage.

Model development. Start with comparable sales and a hedonic/log-price baseline, then evaluate nonlinear models against the same future holdout. Construct comparable-sale features only from transactions available at the decision. Preserve the distinction between homeowner-reported and verified condition, including when each became available. A later inspection cannot retroactively justify an earlier quote.

Validation. Report dollar and relative error, directional bias, tail loss, interval coverage, and performance by geography, price range, condition, and market period. Evaluate acquired inventory separately: good accuracy across all prospects does not imply good accuracy on the selected purchased pool. Nominal interval coverage is not a guarantee under temporal or geographic shift.

The prototype illustrates this with a future-period hedonic-model MAE of **\$25,384**, versus **\$32,384** for its boosted alternative. Nominal 90% intervals cover **86.21%** of future observations. Complexity does not automatically improve forecasting, and uncertainty calibration needs monitoring. These numbers describe the baseline synthetic experiment only.

Implementation boundary. The current code fits point predictions and split-conformal intervals; it does not estimate a fully calibrated joint distribution of exit value, repairs, time, and market shocks. That richer distribution is an architectural objective, not an implemented capability.

Public offer descriptions identify comparable sales, market conditions, verified condition, and human review as relevant inputs [1]. They do not establish a particular model architecture or forecasting accuracy.

3. Acquisition: initial offer, seller signal, escalation

The company's buy decision acts on the homeowner's willingness to sell. Three quantities should remain separate: the owner's belief about market value, the property's market value, and the owner's reservation utility. The reservation utility also reflects outside options, transaction costs, convenience, certainty, and attachment. A refusal does not by itself prove the valuation model is wrong; an acceptance does not validate the quote as market value.

A useful behavioral representation is

$$A(O) = \mathbf{1}\{N(O) + B_{\text{convenience}} \geq R + \varepsilon\},$$

where O is the gross offer, $N(O)$ the seller's net proceeds, and R reservation utility expressed in monetary units. Completed acquisition is a separate event after acceptance. The distinction matters whenever deals cancel or closing conditions change.

Initial offer. Use the valuation distribution, expected costs, and capital/risk limits to choose a feasible quote.

Learning its causal effect requires logging all offered opportunities, not only accepted transactions. The target may be completed acquisitions or expected contribution per original prospect, with a specified closing horizon.

Seller signal. Initial acceptance, continued engagement, a counteroffer, verified condition, and stated expectations can provide additional information. However, these signals are partly responses to the initial quote. Conditioning on them when estimating the initial offer's total effect can remove an important causal pathway or introduce selection bias.

Upward escalation. Once the seller signal has been observed, it becomes part of the history for the next decision. A stage-specific question is the effect of an incremental offer increase among initially unaccepted, still-engaged prospects. Its estimand is narrower than the effect of increasing every initial offer. If the upstream policy changes, the escalation population can change too.

The acquisition objective is a joint expectation:

$$J_{\text{buy}}(a \mid H) = [\mathbf{E}(a) \{G^{\pi_{\text{sell}}}(a) - K(a) - R_{\text{cost}}(a)\} \mid H] - c_{\text{followup}}.$$

H is observed pre-escalation history; C is completed acquisition; K company outlay; R_{cost} actual repairs; and $G^{\pi_{\text{sell}}}$ downstream net disposition after selling and carrying costs under the resale policy.

Increasing the offer changes both the acquisition rate and **which homes are acquired**. Consequently, completion probability multiplied by an unconditional average acquired-home margin is generally insufficient. Historical mature rewards on all eligible prospects can instead support learning the joint objective directly.

The housing-intermediation literature motivates this valuation/selection trade-off [2]. Work on homeowner expectations also helps explain why beliefs affect selling behavior without being interchangeable with a property's physical value [3]. Neither supplies this prototype's response coefficients.

4. Resale: initial list, buyer signal, markdown

The sell decision acts on buyers, so its usual response direction differs from acquisition: a lower list price can increase buyer interest and completion, but can also reduce transaction proceeds.

Initial list. Combine the exit-value distribution with expected response, time to close, negotiation, and holding costs. List price is an intervention, not the target label for a market-value model. An initial-listing experiment should

define its population and follow-up from listing, rather than treating it as the same experiment as repricing already-unsold inventory.

Buyer signal. Views, tours, offers, elapsed time, and comparable-market activity update the decision state. Weak engagement may reflect an excessive price, weak demand, property condition, marketing, or a combination. Historical markdowns are therefore often concentrated on harder-to-sell inventory. A negative historical association between cuts and completion is not evidence that cuts causally reduce completion.

Downward markdown. At a defined checkpoint, estimate effects among homes still eligible and unsold. Use signals observed before that checkpoint. Post-markdown views, offers, or negotiations are mediators/outcomes, not pretreatment controls for the markdown’s total effect.

Let $q_b(H)$ be the probability of completed resale by horizon h under markdown b . A continuation-aware objective is

$$J_{\text{sell}}(b \mid H) = q_b(H) E[U_{\text{early}} \mid \text{close}, b, H] \\ + (1 - q_b(H)) E[U_{\text{continue}} \mid \text{no close}, b, H].$$

The two U terms are net future disposition values, including relevant selling and carrying costs. Acquisition basis is already sunk for incremental repricing comparisons, but must still be charged once when reporting lifecycle profit. A home unsold at the measurement horizon retains an asset value.

The prototype uses a fixed terminal sale 45 days after the repricing decision for clarity. A production system would need a defensible continuation policy and richer timing/cancellation models. Repeated markdowns create a sequential decision problem: effects conditional on historical prices cannot simply be reused as if future inventory composition and buyer signals were unchanged.

5. Causal design across both sides

Valuation is primarily a forecasting task. Causal inference identifies how actions built on that forecast change response, selection, proceeds, and time. It also informs underwriting by revealing the economics of selected inventory; it does not turn observed accepted offers into unbiased market-value labels.

Design element	Acquisition	Resale
Decision population	Original offered prospects, or a separately defined engaged-nonacceptor population	Newly listed inventory, or a separately defined unsold-at-checkpoint population
Action	Initial quote or incremental upward escalation	Initial list or incremental downward markdown
Response	Acceptance and completed acquisition, separately	Interest, offer/contract, and completed resale, separately
Economic outcome	Contribution per eligible prospect, including nonacquisitions	Remaining value and lifecycle contribution, including unsold inventory
Main selection issue	Who requests, remains engaged, accepts, and closes	Who was acquired and remains unsold at the checkpoint

Prefer experiments over operationally feasible actions, with logged assignment probabilities and delivered terms. Randomize within predecision constraints, such as offer caps and list-price floors. A contrast is identified only where both actions have support; a constrained cohort cannot support a universal recommendation outside its feasible range.

Use outcome regression for precision and doubly robust estimation to combine predictions with propensity-weighted residual corrections. Cross-fitting separates nuisance estimation from scoring. Under observational assignment, this still requires adequate overlap and no unmeasured confounding; flexible machine learning does not supply those assumptions.

Instrumental variables: where credible assignment affects a delivered price but not outcomes through other pathways, instrument-based methods may identify a narrower causal response. They require additional design assumptions. Detailed instrument selection and diagnostics are outside this summary. The linked prototype randomizes fully delivered actions and does not need this additional machinery.

Keep policy development separate from final evaluation. Estimate differences between fixed policies on the same held-out prospects, with uncertainty on the paired differences. Do not infer superiority by comparing two independent intervals or by selecting the most favorable result from many policies. Missing or immature outcomes are not negative labels.

6. Connect the stages through downstream value

The implemented extension follows stable prospect IDs through acquisition, listing, markdown, and disposition. It has one escalation and one markdown; the initial-offer and initial-list mechanisms are fixed upstream inputs.

Stage	Implemented contract
Acquisition escalation	Day 7 after the initial quote, for engaged nonacceptors; feasible increases of 0/1/2/3%
Acquisition outcomes	Acceptance within 14 days and completed acquisition within 35 days of escalation
Resale markdown	Listing day 21, conditional on acquisition, no early sale, and weak buyer signal; feasible cuts of 0/1/2/3%
Resale outcomes	Completed resale over the next 30 days; otherwise terminal disposition 45 days after the markdown checkpoint
Complete reward	Observed through offer day 150; company outlay, repairs, selling, and carrying accounted for once

Three chronological cohorts separate learning and evaluation:

1. **Development: 12,000 prospects.** Fit stage responses and define a resale rule from observable information.
2. **Continuation learning: 10,000 prospects.** Use mature downstream outcomes and known resale propensities to construct policy-adjusted rewards; fit acquisition joint values on all escalation-eligible prospects.
3. **Evaluation: 8,000 prospects.** Freeze the learned rules, apply sequential doubly robust scores, and compare lifecycle contribution.

Every earlier cohort's lifecycle outcomes mature before the next cohort begins. The shared AVM is fitted on separate earlier market transactions. At each eligible resale decision, the downstream score combines predicted value under the target action with an inverse-propensity correction from the logged action. The acquisition score applies the same correction to the resulting lifecycle reward. This connects downstream policy to acquisition learning without using future outcomes as decision inputs [4,5].

The implementation uses allowlisted features, as-of forecast margin caps, known conditional action support, and separate evaluator-only truth. The acquisition rule selects the smallest near-optimal feasible increase whose **own predicted gain** meets a \$150 hurdle. A maximum-candidate gain cannot be reported as the gain of a different chosen action. Holding every offer is a valid outcome.

7. Portfolio economics, risk, and capital

Individual conversion and margin curves are inputs to a portfolio decision, not its final objective. A coherent formulation is

$$\max_{\pi_{\text{buy}}, \pi_{\text{sell}}} E \left[\sum_i \Pi_i \right] - \lambda \mathcal{R} \left(\sum_i \Pi_i \right)$$

subject to capital, concentration, service, and action constraints .

Π_i is lifecycle contribution and $\mathcal{R}$ a specified downside-risk measure. This is a **proposed general formulation**, not a claim that the current prototype implements a full capital/risk optimizer.

Layer	Required discipline
Accounting	Separate seller net proceeds, company outlay, actual repairs, selling, holding, and financing; charge each component once
Capital	Model cash deployed over time, not only average completed acquisitions; cancellation and holding-time uncertainty affect exposure
Portfolio risk	Preserve shared market shocks and concentration; independent-home uncertainty understates correlated downside
Objective	Distinguish contribution dollars, gross profit, and margin percentage; maximizing the ratio can sacrifice scale
Constraints	Use supported actions and explicit business limits; fitted expected-volume feasibility is not a hard realized guarantee

In the **illustrative ledger**, a \$450,000 gross offer with a 2% retained charge and \$6,000 repair credit gives \$435,000 company outlay. A separate \$3,000 seller closing expense leaves \$432,000 seller net. Actual repair spending remains another cost. These are synthetic assumptions, not a statement of current company fees; public documentation describes variable charges and distinct offer components [6].

The existing portfolio example trades expected completion against contribution using lotteries over tested prices. It does not interpolate an unsupported average quote. It also does not enforce realized cash-path limits, jointly optimize both sides' capital, or solve a full dynamic inventory-control problem.

In its synthetic reference frontier, increasing fitted expected completion from **32.46% to 42.56%** reduces fitted contribution from roughly **\$40,485 to \$38,975 per home**. This is an economic trade-off under model estimates, not a forecast guarantee or a capital-risk stress result.

8. Illustrative evidence and interpretation

The results below come from the saved seed-42 experiments. They demonstrate failure modes and decision trade-offs, not real-market policy effectiveness. The baseline and linked experiment have different populations and assumptions.

Finding	Interpretation
Baseline randomized resale completion lift: +8.95 percentage points; naive observational difference: -5.80 points	Operational selection into cuts can reverse the historical association
One baseline home's 3% cut raises fitted completion from 17.86% to 25.93%, while contribution falls from \$32,358 to \$30,559	A positive response effect does not imply an economically attractive action
Linked acquisition +3%: estimated completion lift +3.52 points, 95% interval [0.88, 6.17], matching oracle +3.87 points	Stage-specific effects concern completed acquisition within the defined eligible population
Linked resale 3% cut: estimated lift +21.06 points, interval [11.66, 30.46], matching oracle +8.96 points	This interval misses its target; finite-sample uncertainty and model error remain material
Linked selective rule holds all 4,214 escalation-eligible offers; estimated value difference vs hold-both is -\$274 per eligible prospect, interval [-\$1,288, \$740]	The experiment does not establish a joint-policy profit improvement; the rule must be allowed not to escalate

The matching acquisition contrast uses 2,457 prospects with support for every action; the resale contrast uses 750 markdown-eligible homes. These are not the same population or one combined effect. Oracle values are known only inside the simulator and never supply decision features.

For all 8,000 original linked prospects, including initial acceptors, the selective-rule difference is approximately +\$5 per prospect, with interval [-\$713, \$723]. The denominator and affected paths differ from the eligible-only comparison. Absolute modeled contribution is negative under this fixed upstream system; downstream optimization does not establish business viability.

The policy intervals are pointwise and conditional on frozen learned rules. They do not incorporate every form of policy search, shared market uncertainty, or parameter drift. Unfavorable results are retained rather than selecting parameters or seeds to manufacture a winning policy.

9. Implementation and operating priorities

The prototype shares one Python implementation across command-line execution, generated reports, examples, and a local interactive application. It keeps the baseline experiment and the linked extension separate.

Component	Implemented responsibility
features.py, valuation.py	Point-in-time comparable sales, model selection, chronological evaluation, and intervals
seller.py, causal.py	Offer accounting, homeowner response/selection examples, and causal response estimation
optimization.py, evaluation.py	Continuation-aware economics, supported policies, portfolio completion trade-offs, and evaluation
two_sided_simulation.py, two_sided.py	Linked trajectories, mature reward learning, action constraints, and sequential policy evaluation

Component	Implemented responsibility
pipeline.py, CLI, examples, app	Reproducible artifacts and shared presentation, without a separate dashboard-only model

Running the studies creates reference artifacts in `outputs\reference` and `outputs\two_sided`; datasets are not included in a fresh checkout. The maintained summary, contracts, sources, and generated result snapshots remain in docs. This summary does not reproduce every diagnostic.

```
python -m home_valuation pipeline --seed 42 --output outputs\reference
python -m home_valuation two-sided --seed 42 --output outputs\two_sided
python -m streamlit run app\streamlit_app.py
```

Before operational use, prioritize four workstreams:

- 1. **Measurement and logging.** Capture original risk sets, action assignments, delivered terms, signals, timestamps, overrides, cancellations, and complete cash flows, including nonconverting opportunities.
- 2. **Causal learning.** Validate feasible stage experiments; evaluate changes to initial policies separately; handle treatment-induced signals, shifting eligibility, and dependence across properties and markets.
- 3. **Risk integration.** Calibrate joint value/cost/time scenarios and connect both sides to capital exposure, concentration limits, and explicit downside objectives.
- 4. **Monitoring and governance.** Track value residuals, selected-inventory error, completion calibration, action support, reward maturity, and realized economics. Review customer treatment and legal obligations independently of model accuracy.

The organizing principle is stable across implementations: **forecast asset value, identify how actions change behavior and selection, and allocate capital using the resulting distribution of lifecycle outcomes.** A valuation, a purchase offer, and a resale list price remain distinct throughout.

Selected references

These sources provide context and methodology, not empirical calibration of the synthetic results. Detailed verification limits are recorded in the separate source ledger.

- 1. [Opendoor: How does Opendoor determine my offer price?](#) Public valuation inputs and assessment context.
- 2. [Buchak et al.: Why is Intermediating Houses so Difficult? Evidence from iBuyers](#) Housing intermediation, valuation, and selection.
- 3. [Bottan and Perez-Truglia: Betting on the House](#) Subjective expectations and selling behavior; not an offer-price elasticity.
- 4. [Dudik, Langford, and Li: Doubly Robust Policy Evaluation and Learning](#) Combining response prediction with propensity correction.
- 5. [Jiang and Li: Doubly Robust Off-policy Value Evaluation for Reinforcement Learning](#) Extension of policy evaluation to sequential decisions.
- 6. [Opendoor: What’s included in my offer?](#) Offer components, charges, and seller net proceeds.